K. Manideep Sai

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B.Tech Artificial Intelligence & Machine Learning student with hands-on experience building AI/ML applications, REST APIs, browser extensions, and full-stack projects using Python, FastAPI, JavaScript, React, and Node.js. Experienced in integrating machine learning models, LLMs, and external APIs into practical applications, with a strong interest in problem solving, backend development, and Generative AI.

EDUCATION

CVR College of Engineering — Expected Graduation: 2027
Bachelor of Technology in Artificial Intelligence & Machine Learning | *CGPA: 8.17* — Hyderabad, India

Narayana Jr College — 2023
Intermediate (Class 12), TS State Board | *Marks: 96.5%* — Hyderabad, India

Narayana e-Techno School — 2021
Secondary School Certificate (Class 10) | *GPA: 10.0/10.0* — Hyderabad, India

PROJECTS

HealBuddy | **AI-Assisted Health Information Platform** | *FastAPI, Python, Scikit-learn, Ollama, HTML/JS*

- Built an interactive health information application utilizing a RandomForestClassifier to predict possible conditions based on user-provided symptoms.
- Developed a FastAPI backend to connect the web frontend with the machine-learning inference pipeline.
- Integrated a locally hosted Llama 3.2 3B model via Ollama to generate conversational explanations and contextual health guidance.

SaralQ | **Academic Translation & Learning Extension** | *JavaScript, Node.js, Express, Gemini Flash 2*

- Developed a Chromium browser extension for one-click translation of research papers and web pages, integrating external Bhashini API services.
- Added language selection and AI-generated multimedia features (summaries and quizzes) utilizing the Gemini Flash 2 API.
- Implemented localized caching for translated content to optimize performance and prevent redundant API translation requests.
- Built Node.js/Express backend components to mediate communication between the browser extension and external APIs.

Ingreedy | **Predictive Recipe Recommendation Engine** | *Python, React, MongoDB, Gemini API*

- Built a full-stack recipe recommendation application that matches user-provided ingredients against a large dataset using NLP-based retrieval (TF-IDF and Cosine Similarity).
- Integrated the Gemini API to serve as a conversational interface, enhancing retrieved recipes with dynamic explanations and substitutions.
- Implemented YouTube API search integration to surface relevant video demonstrations for suggested recipes.

ADDITIONAL TRAINING

AI/ML Virtual Internship (10 Weeks) — 2025
AICTE & EduSkills (Supported by Google) — Remote

- Gained practical exposure to data preparation, feature engineering, and model training workflows.
- Participated in evaluating machine learning models and understanding fundamental data science pipelines.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL
CS & Core Concepts: Data Structures & Algorithms, Object-Oriented Programming (OOP), Problem Solving
AI & Machine Learning: Scikit-learn, Pandas, NumPy, NLP, LLM Integration, Ollama, Gemini API, Prompt Eng.
Backend & Web: FastAPI, REST APIs, Node.js, Express, React, HTML/CSS
Databases & Tools: PostgreSQL, MongoDB, Oracle SQL, Git/GitHub, Linux

ACHIEVEMENTS & CERTIFICATIONS

- **Hackathons & Competitions:** Megathon Finalist (2025) – presented SaralQ; Agentathon 2025 Participant; 1st Place Smart India Hackathon (SIH) Internal Selections (2024).
- **Cloud & Database Certifications:** Oracle Certified Database Programming with SQL, AWS Academy Cloud Foundations.
- **Programming & AI Certifications:** Python Programming with Red Hat, Honeywell Future Skills Edge on AI, NPTEL Joy of Computing with Python.